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Division: A2

Subject: Programming with Java

ASSIGNMENT No : 8

TITLE :Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement

Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives

To demonstrate the use of the super keyword for accessing parent class members.

Theory

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class

constructors, access parent class variables, and call overridden methods. Super helps distinguish

between parent and child class members with the same name. It ensures proper constructor chaining

during object creation. It is an important concept in inheritance and method overriding.

Outcome

Conclusion

This program demonstrates the use of the super **keyword** to access the parent constructor from the child class. This program demonstrates the use of the super **keyword** to initialize parent class data in the child class.

Exercise

1. Create an Employee and Manager program where Manager accesses Employee details using super keyword.

```
import java.util.Scanner;
```

```
class Employee {  
    int id;  
    String name;  
  
    Employee(int id, String name) {  
        this.id = id;  
        this.name = name;  
    }  
}  
  
class Manager extends Employee {  
    double salary;  
  
    Manager(int id, String name, double salary) {  
        super(id, name); // Access parent constructor  
        this.salary = salary;  
    }  
  
    void display() {  
        System.out.println("ID: " + id);  
        System.out.println("Name: " + name);  
        System.out.println("Salary: " + salary);  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);

System.out.print("Enter ID: ");
int id = sc.nextInt();
sc.nextLine();

System.out.print("Enter Name: ");
String name = sc.nextLine();

System.out.print("Enter Salary: ");
double salary = sc.nextDouble();

Manager m = new Manager(id, name, salary);
m.display();
}
}
```

2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.

```
import java.util.Scanner;

class Vehicle {
    String vehicleNo;
    String owner;

    Vehicle(String vehicleNo, String owner) {
        this.vehicleNo = vehicleNo;
        this.owner = owner;
    }
}
```

```
}
```

```
class CarInsurance extends Vehicle {
```

```
    double premium;
```

```
    CarInsurance(String vehicleNo, String owner, double premium) {
```

```
        super(vehicleNo, owner); // Access parent constructor
```

```
        this.premium = premium;
```

```
    }
```

```
    void display() {
```

```
        System.out.println("Vehicle No: " + vehicleNo);
```

```
        System.out.println("Owner: " + owner);
```

```
        System.out.println("Premium: " + premium);
```

```
    }
```

```
}
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter Vehicle Number: ");
```

```
        String no = sc.nextLine();
```

```
        System.out.print("Enter Owner Name: ");
```

```
        String owner = sc.nextLine();
```

```
        System.out.print("Enter Premium: ");
```

```
double premium = sc.nextDouble();
```

```
CarInsurance c = new CarInsurance(no, owner, premium);
```

```
c.display();
```

```
}
```

```
}
```